

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 06 · Geometric Sequences

With Answers

Questions with full worked solutions

- 20 pages
- Past papers up to January 2026

PURE 2 | TOPIC 06

Geometric Sequences

Your classified route

7

QUESTIONS

55

MARKS

3

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 06 | FAMILY 01

Infinite Sum

WHAT THIS FAMILY ASKS

Recognise the infinite sum demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 State the convergence condition before using the formula.

3 classified questions

24 total marks

T06

06_INFTY

06_INFTY_CORE

CHAPTER
Geometric Sequences

FAMILY
Infinite sum

VARIANT
core family move

8 marks

WMA12/01 MAY/JUNE 2022

Question 6

Past-paper question

WMA12-01_2022_MayJune_Q06

6. In a geometric sequence $u_1, u_2, u_3, \dots$

- the common ratio is r
- $u_2 + u_3 = 6$
- $u_4 = 8$

(a) Show that r satisfies

$$3r^2 - 4r - 4 = 0$$

(3)

Given that the geometric sequence has a sum to infinity,

(b) find u_1

(3)

(c) find S_∞

(2)

T06 Geometric Sequences

06_INFTY Infinite sum

06_INFTY_CORE core family
move

WORKED SOLUTION

Question 6

WMA12/01 May/June 2022

WMA12-01_2022_MayJune_Q06

1

Write the terms

$$u_2 = u_1 r, u_3 = u_1 r^2, \text{ and } u_4 = u_1 r^3.$$

2

Use u four

$$\text{Since } u_4 = 8, u_1 r^3 = 8, \text{ so } u_1 r = \frac{8}{r^2}.$$

3

Use u two plus u three

$$u_2 + u_3 = u_1 r(1 + r) = 6.$$

4

Show the quadratic

$$\text{Substitute } u_1 r = \frac{8}{r^2} \text{ to get } \frac{8}{r^2}(1 + r) = 6. \text{ Hence } 8 + 8r = 6r^2, \text{ so } 3r^2 - 4r - 4 = 0.$$

5

Solve for r

$$3r^2 - 4r - 4 = 0 \text{ gives } r = 2 \text{ or } r = -\frac{2}{3}.$$

6

Choose the sum-to-infinity ratio

$$\text{A sum to infinity requires } |r| < 1, \text{ so } r = -\frac{2}{3}.$$

7

Find u one

$$u_1 r^3 = 8, \text{ so } u_1 \left(-\frac{2}{3}\right)^3 = 8 \text{ and } u_1 = -27.$$

8

Find the infinite sum

$$S_\infty = \frac{u_1}{1-r} = \frac{-27}{1+2/3} = -\frac{81}{5}.$$

FINAL ANSWER

WMA12-01_2022_MayJune_Q06

$$(a) \text{ Proven: } 3r^2 - 4r - 4 = 0. \quad (b) u_1 = -27. \quad (c) S_\infty = -\frac{81}{5}.$$

T06

06_INFTY

06_INFTY_CORE

CHAPTER
Geometric Sequences

FAMILY
Infinite sum

VARIANT
core family move

6 marks

WMA12/01 JANUARY 2025

Question 4

Past-paper question

WMA12-01_2025_Jan_Q04

4. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Given that, in a particular geometric series,

- the sum of the first three terms is 70.2
- the sum to infinity is 75

find, for this series,

(a) the common ratio,

(4)

(b) the first term.

(2)

T06 Geometric Sequences

06_INFTY Infinite sum

06_INFTY_CORE core family
move

WORKED SOLUTION

Question 4

WMA12/01 January 2025

WMA12-01_2025_Jan_Q04

1

Write the two sums

For first term a and common ratio r , $S_{\infty} = \frac{a}{1-r} = 75$ and $S_3 = \frac{a(1-r^3)}{1-r} = 70.2$.

2

Divide the equations

$\frac{S_3}{S_{\infty}} = 1 - r^3$, so $\frac{70.2}{75} = 1 - r^3$.

3

Find the ratio

$0.936 = 1 - r^3$, so $r^3 = 0.064$ and $r = 0.4$.

4

Find the first term

Use $\frac{a}{1-0.4} = 75$. Hence $a = 75(0.6) = 45$.

FINAL ANSWER

(a) $r = 0.4$. (b) 45.

WMA12-01_2025_Jan_Q04

T06

06_INFTY

06_INFTY_HV01

CHAPTER
Geometric Sequences

FAMILY
Infinite sum

VARIANT
State the convergence
condition before using the
formula.

10 marks

WMA12/01 OCTOBER 2021

Question 7

Past-paper question

WMA12-01_2021_Oct_Q07

7. (i) A geometric sequence has first term 4 and common ratio 6
Given that the n^{th} term is greater than 10^{100} , find the minimum possible value of n .
(3)

- (ii) A different geometric sequence has first term a and common ratio r .

Given that

- the second term of the sequence is -6
- the sum to infinity of the series is 25

- (a) show that

$$25r^2 - 25r - 6 = 0 \quad (3)$$

- (b) Write down the solutions of

$$25r^2 - 25r - 6 = 0 \quad (1)$$

Hence,

- (c) state the value of r , giving a reason for your answer,
(1)

- (d) find the sum of the first 4 terms of the series.
(2)

T06 Geometric Sequences

06_INFTY Infinite sum

06_INFTY_HV01 State the convergence condition before using the formula.

WORKED SOLUTION

Question 7

WMA12/01 October 2021

WMA12-01_2021_Oct_Q07

1

Write the n th term

The n th term of the first sequence is $4 \cdot 6^{n-1}$.

2

Use logarithms

We need $4 \cdot 6^{n-1} > 10^{100}$, so $n - 1 > \frac{100 - \log_{10} 4}{\log_{10} 6} = 127.736 \dots$

3

Choose the smallest integer

Thus n must be greater than 128.736..., so the minimum possible value is 129.

4

Use the second term

For the second sequence, $ar = -6$, so $a = -\frac{6}{r}$.

5

Use the sum to infinity

$\frac{a}{1-r} = 25$. Substituting $a = -\frac{6}{r}$ gives $-6 = 25r(1-r)$.

6

Show the quadratic

Rearranging gives $25r^2 - 25r - 6 = 0$.

7

Solve for r

$25r^2 - 25r - 6 = 0$ gives $r = \frac{6}{5}$ or $r = -\frac{1}{5}$.

8

Choose the valid ratio

A sum to infinity exists only when $|r| < 1$, so $r = -\frac{1}{5}$.

9

Find the first term

Since $ar = -6$, $a = 30$.

10

Sum four terms

$S_4 = 30 \frac{1 - (-1/5)^4}{1 - (-1/5)} = \frac{624}{25}$.

FINAL ANSWER

WMA12-01_2021_Oct_Q07

(i) $n = 129$. (ii)(a) **Proven.** (ii)(b) $r = \frac{6}{5}$ **or** $-\frac{1}{5}$. (ii)(c) $r = -\frac{1}{5}$. (ii)(d) $\frac{624}{25}$.

DR ESLAM AHMED

TOPIC 06 | FAMILY 02

Common Ratio

WHAT THIS FAMILY ASKS

Recognise the common ratio demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**1 classified questions**

7 total marks

T06

06_RATIO

06_RATIO_CORE

CHAPTER
Geometric Sequences

FAMILY
Common ratio

VARIANT
core family move

7 marks

WMA12/01 OCTOBER 2022
Question 8

Past-paper question

WMA12-01_2022_Oct_Q08

8. A geometric sequence has first term a and common ratio r

Given that $S_{\infty} = 3a$

- (a) show that $r = \frac{2}{3}$

(2)

Given also that

$$u_2 - u_4 = 16$$

where u_k is the k^{th} term of this sequence,

- (b) find the value of S_{10} giving your answer to one decimal place.

(5)

T06 Geometric Sequences

06_RATIO Common ratio

06_RATIO_CORE core family
move

WORKED SOLUTION

Question 8

WMA12/01 October 2022

WMA12-01_2022_Oct_Q08

1

Use the sum to infinity formulaFor a geometric sequence, $S_{\infty} = \frac{a}{1-r}$. The question gives $S_{\infty} = 3a$.

2

Find the common ratio $\frac{a}{1-r} = 3a$. Since $u_2 - u_4 = 16$, $a \neq 0$, so $\frac{1}{1-r} = 3$ and $r = \frac{2}{3}$.

3

Use u_2 minus u_4 $u_2 - u_4 = ar - ar^3 = ar(1 - r^2) = 16$.

4

Substitute r $a\left(\frac{2}{3}\right)\left(1 - \frac{4}{9}\right) = 16$.

5

Solve for a $a \cdot \frac{2}{3} \cdot \frac{5}{9} = 16$, so $\frac{10a}{27} = 16$ and $a = \frac{216}{5} = 43.2$.

6

Find S_{10} $S_{10} = \frac{a(1-r^{10})}{1-r} = 3a\left(1 - \left(\frac{2}{3}\right)^{10}\right)$.

7

Evaluate $S_{10} = 3(216/5)(1 - (2/3)^{10}) = 92840/729 = 127.352\dots$, so $S_{10} = 127.4$ to 1 decimal place, as requested in the question.**FINAL ANSWER**

WMA12-01_2022_Oct_Q08

(a) $r = 2/3$. (b) $a = 43.2$; $S_{10} = 92840/729 = 127.352\dots$, so $S_{10} = 127.4$ to 1 d.p.

TOPIC 06 | FAMILY 03

Finite Sum

WHAT THIS FAMILY ASKS

Recognise the finite sum demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 If a sum is given, solve the resulting exponential equation carefully.

3 classified questions

24 total marks

T06

06_SUM

06_SUM_CORE

CHAPTER
Geometric Sequences

FAMILY
Finite sum

VARIANT
core family move

8 marks

WMA12/01 OCTOBER 2025

Question 8

Past-paper question

WMA12-01_2025_Oct_Q08

8. (i) A geometric series has first term a and common ratio r .

Prove that the sum of the first n terms of this series S_n is given by

$$S_n = \frac{a(1-r^n)}{1-r} \quad (3)$$

- (ii) A liquid is to be stored in a barrel.

Due to evaporation, the volume of the liquid in the barrel at the end of each year is 8% less than the volume of the liquid in the barrel at the start of the year.

At the start of the first year, the barrel is filled with 150 litres of the liquid.

- (a) Show that the amount of the liquid in the barrel at the end of 6 years is approximately 91 litres. (2)

At the start of each year a new barrel is filled with 150 litres of the liquid so that, at the end of 40 years, there are 40 barrels containing the liquid.

- (b) Calculate the total amount of the liquid, to the nearest litre, in the 40 barrels at the end of 40 years. (3)

T06 Geometric Sequences

06_SUM Finite sum

06_SUM_CORE core family
move

WORKED SOLUTION

Question 8

WMA12/01 October 2025

WMA12-01_2025_Oct_Q08

1

Write S_n

$$S_n = a + ar + ar^2 + \dots + ar^{n-1}.$$

2

Multiply by r

$$rS_n = ar + ar^2 + ar^3 + \dots + ar^n.$$

3

Subtract

$$S_n - rS_n = a - ar^n = a(1 - r^n).$$

4

Rearrange

$$(1 - r)S_n = a(1 - r^n), \text{ so } S_n = \frac{a(1 - r^n)}{1 - r}.$$

5

Use the yearly multiplier

An 8% decrease each year means the multiplier is 0.92.

6

Show the amount after six years

After 6 years, the amount is $150(0.92)^6 = 90.953 \dots$, which is approximately 91 litres.

7

Set up the forty barrel total

At the end of 40 years, the amounts are $150(0.92), 150(0.92)^2, \dots, 150(0.92)^{40}$.

8

Sum the geometric sequence

The total is $150(0.92) \frac{1 - 0.92^{40}}{1 - 0.92} = 1663.581 \dots$

9

Round the total

To the nearest litre, the total amount is 1664 litres.

FINAL ANSWER

WMA12-01_2025_Oct_Q08

(i) Proven. (ii)(a) Approximately 91 litres. (ii)(b) 1664 litres.

T06

06_SUM

06_SUM_HV03

CHAPTER

Geometric Sequences

FAMILY

Finite sum

VARIANT

If a sum is given, solve the
resulting exponential
equation carefully.

7 marks

WMA12/01 JANUARY 2023

Question 7

Past-paper question

WMA12-01_2023_Jan_Q07

7. A geometric sequence has first term a and common ratio r , where $r > 0$

Given that

- the 3rd term is 20
- the 5th term is 12.8

- (a) show that $r = 0.8$

(1)

- (b) Hence find the value of a .

(2)

Given that the sum of the first n terms of this sequence is greater than 156

- (c) find the smallest possible value of n .

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(4)

T06 Geometric Sequences

06_SUM Finite sum

06_SUM_HV03 If a sum is given, solve the resulting exponential equation carefully.

WORKED SOLUTION

Question 7

WMA12/01 January 2023

WMA12-01_2023_Jan_Q07

1

Write the given terms

For a geometric sequence, the third term is $ar^2 = 20$ and the fifth term is $ar^4 = 12.8$.

2

Divide the equations

$$\frac{ar^4}{ar^2} = \frac{12.8}{20}, \text{ so } r^2 = 0.64.$$

3

Use r positive

Since $r > 0$, $r = 0.8$.

4

Find a

Using $ar^2 = 20$, $a(0.8)^2 = 20$, so $a = \frac{20}{0.64} = 31.25$.

5

Write the sum formula

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{31.25(1-0.8^n)}{0.2} = 156.25(1-0.8^n).$$

6

Set up the inequality

We need $156.25(1-0.8^n) > 156$.

7

Isolate the power

$$1 - 0.8^n > \frac{156}{156.25} = \frac{624}{625}, \text{ so } 0.8^n < \frac{1}{625}.$$

8

Use logarithms

$n \log(0.8) < \log\left(\frac{1}{625}\right)$. Since $\log(0.8) < 0$, dividing reverses the inequality: $n > \frac{\log(1/625)}{\log(0.8)} = 28.85 \dots$

9

Choose the smallest integer

The smallest possible integer value is therefore $n = 29$.

FINAL ANSWER

(a) $r = 0.8$. (b) $a = 31.25$. (c) $n = 29$.

WMA12-01_2023_Jan_Q07

T06

06_SUM

06_SUM_HV03

CHAPTER

Geometric Sequences

FAMILY

Finite sum

VARIANT

If a sum is given, solve the
resulting exponential
equation carefully.

9 marks

WMA12/01 MAY/JUNE 2023

Question 6

Past-paper question

WMA12-01_2023_MayJune_Q06

6. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A software developer released an app to download.

The numbers of downloads of the app each month, in thousands, for the first three months after the app was released were

$$2k - 15 \quad k \quad k + 4$$

where k is a constant.

Given that the numbers of downloads each month are modelled as a geometric series,

(a) show that $k^2 - 7k - 60 = 0$ (2)

(b) predict the number of downloads in the 4th month. (4)

The **total** number of all downloads of the app is predicted to exceed 3 million for the first time in the N th month.

(c) Calculate the value of N according to the model. (3)

T06 Geometric Sequences

06_SUM Finite sum

06_SUM_HV03 If a sum is given, solve the resulting exponential equation carefully.

WORKED SOLUTION

Question 6

WMA12/01 May/June 2023

WMA12-01_2023_MayJune_Q06

1

Use the geometric condition

For three consecutive terms of a geometric sequence, the middle term squared equals the product of the outer terms.

2

Set up the equation

So $k^2 = (2k - 15)(k + 4)$.

3

Show the required quadratic

Expanding gives $k^2 = 2k^2 - 7k - 60$, hence $k^2 - 7k - 60 = 0$.

4

Find k

$k^2 - 7k - 60 = (k - 12)(k + 5) = 0$, so $k = 12$ or $k = -5$. The download numbers must be positive, so $k = 12$.

5

Find the common ratio

The first three monthly values are 9, 12 and 16 thousand. The common ratio is $\frac{12}{9} = \frac{4}{3}$.

6

Predict month four

The 4th month is $16 \cdot \frac{4}{3} = \frac{64}{3}$ thousand downloads, which is about 21300 downloads.

7

Write the cumulative total

In thousands, the total after N months is $S_N = \frac{9\left(\left(\frac{4}{3}\right)^N - 1\right)}{\frac{4}{3} - 1} = 27\left(\left(\frac{4}{3}\right)^N - 1\right)$.

8

Compare with three million

Three million downloads is 3000 thousand, so we need $27\left(\left(\frac{4}{3}\right)^N - 1\right) > 3000$.

9

Solve the inequality

$\left(\frac{4}{3}\right)^N > \frac{1009}{9}$. Taking logarithms gives $N > \frac{\log(1009/9)}{\log(4/3)} = 16.405 \dots$

10

Choose the first month

The first integer month for which the total exceeds 3 million is $N = 17$.

FINAL ANSWER

WMA12-01_2023_MayJune_Q06

(a) **Proven.** (b) $\frac{64}{3}$ **thousand downloads, about 21300 downloads.** (c) $N = 17$.

DR ESLAM AHMED